Set Theory

Representations, Subsets, Set Operations, Inclusion–Exclusion, and Set Algebra

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1 Foundations of Set Theory

Set theory is the basic language of discrete mathematics and computer science. Whether we are talking about a database table, the vertices of a network graph, a formal language, or a sample space in probability, we are working with sets.

DEFINITION **Set**

A **set** is an unordered collection of distinct objects, called the **elements** or **members** of the set. A set is said to **contain** its elements:

- $a \in A$ denotes that a is an element of set A.
- $a \notin A$ denotes that a is **not** an element of set A.

By convention, uppercase letters ($A, B, C, \ldots$) denote sets and lowercase letters ($a, b, c, \ldots$) denote individual elements:

- $A = \{2, 4, 6, 8\}$ is a finite set of four even integers ($2 \in A$, whereas $3 \notin A$).
- $B = \{a, b, c, \ldots, x, y, z\}$ denotes the lowercase letters of the English alphabet, where the ellipsis ($\ldots$) indicates a clear pattern.
- $\boldsymbol{P}$ denotes the infinite set of all positive integers.

INTUITION & DEEP DIVE **Why 'Unordered' and 'Distinct' Matter**

In programming, a mathematical set corresponds to a `Set` data structure (like `set` in Python or `std::unordered_set` in C++), rather than a `List` or `Array`:

1. **Unordered:** Membership ($a \in A$) does not depend on position; $\{2, 4, 6, 8\}$ and $\{8, 2, 6, 4\}$ are the exact same set.
2. **No Duplicates:** Listing an element more than once does not change the set ($\{1, 2, 2, 3\} = \{1, 2, 3\}$). When multiplicity matters, we use **multisets** (bags); when order matters, we use **tuples** or **sequences**.

2 Representations of Sets

We can describe a set in four main ways depending on whether it is finite, defined by a rule, or one of the standard number systems.

2.1 Roster (Enumeration) Method

When a set is finite (or has an obvious pattern), we can list its members inside curly braces `{ }`:

- $A = \{1, 2, 3, 4\}$ — a set of four integers.
- $B = \{a, b, \{a, b, c\}, \{a, c\}\}$ — a set with four elements, two of which are themselves sets ($a \in B$ and $\{a, c\} \in B$, though $c \notin B$).
- $H = \{\text{dog}, \text{cat}, \text{crocodile}\}$ — a set of three animals.

2.2 Standard Number Systems

Several number sets come up so often that they have dedicated boldface symbols:

Symbol	Name	Definition / Elements
$\boldsymbol{P}$	Positive integers	$\{1, 2, 3, \ldots\}$

$\boldsymbol{N}$	Natural numbers	$\{1, 2, \ldots\}$ (or $\{0, 1, 2, \ldots\}$ depending on convention)
$\boldsymbol{Z}$	Integers	$\{\ldots, -2, -1, 0, 1, 2, \ldots\}$
$\boldsymbol{Q}$	Rational numbers	$\left\{\frac{p}{q} \mid p, q \in \boldsymbol{Z}, q \neq 0\right\}$
$\boldsymbol{R}$	Real numbers	All rational and irrational numbers on the number line
$\boldsymbol{C}$	Complex numbers	$\{a + bi \mid a, b \in \boldsymbol{R}, i^2 = -1\}$

PITFALL / WARNING **Does N Start at 0 or 1?**

Different authors disagree on whether 0 is a natural number. Here we follow the convention $\boldsymbol{N} = \{1, 2, 3, \ldots\}$ (matching $\boldsymbol{P}$), though many textbooks write $\boldsymbol{N} = \{0, 1, 2, \ldots\}$ and $\boldsymbol{Z}^+$ for $\{1, 2, 3, \ldots\}$. It is always good practice to check which convention a problem uses when 0 is a boundary case.

2.3 Set-Builder Notation

When listing every element is impractical, we describe a set by stating the condition an object x must satisfy to belong to it:

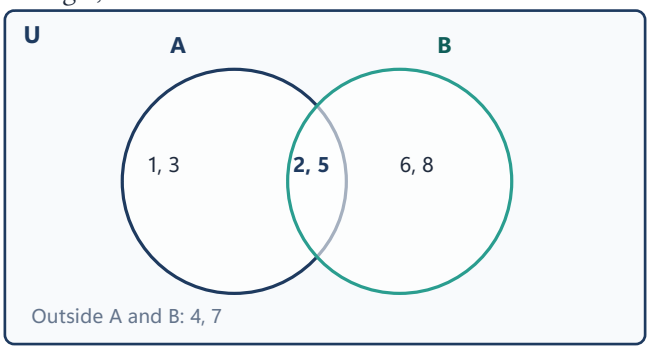
$$S = \{x \mid \text{condition on } \ x\}$$

The vertical bar $\mid$ is read as **"such that"**.

- **Example 1:** $A = \{x \mid x \in \boldsymbol{P}, x < 5\}$ gives $A = \{1, 2, 3, 4\}$.
- **Example 2:** $B = \left\{x \mid \frac{x}{2} \in \boldsymbol{P}, 2 \leq x \leq 8\right\}$ requires x to be a positive even integer between 2 and 8, so $B = \{2, 4, 6, 8\}$.
- **Example 3:** $D = \{x \mid x \in \boldsymbol{N}, x < 5 \vee x > 10\}$ gives the infinite set $D = \{1, 2, 3, 4, 11, 12, 13, \ldots\}$.

2.4 Venn Diagrams

A **Venn diagram** represents sets visually. The **universal set** U—which contains all objects under consideration—is drawn as a bounding rectangle, and subsets of U are drawn as circles inside it.



Venn diagram for U = {1, 2, ..., 8}, A = {1, 2, 3, 5}, and B = {2, 5, 6, 8}.

EXAMPLE & WALKTHROUGH **Roster and Set-Builder Practice**

1. List the members of each set:

- **(a)** $\{x \mid x$ is a real number such that $x^2 = 1\}$: Solving $x^2 = 1$ over $\boldsymbol{R}$ gives $\boldsymbol{\{-1, 1\}}$.
- **(b)** $\{x \mid x$ is a positive integer less than $12\}$: $\boldsymbol{\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}}$.
- **(c)** $\{x \mid x$ is the square of an integer and $x < 100\}$: The integers whose squares are under 100 are $-9, \ldots, 9$, producing the squares $\boldsymbol{\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}}$.
- **(d)** $\{x \mid x$ is an integer such that $x^2 = 2\}$: Since $\pm\sqrt{2}$ are not integers, no integer satisfies this condition: $\varnothing$ (or $\boldsymbol{\{\}}$).

2. Describe each set using set-builder notation:

- **(a)** $\{0, 3, 6, 9, 12\}$: Multiples of 3 from 0 to 12: $\boldsymbol{\{3k \mid k \in \boldsymbol{Z}, 0 \leq k \leq 4\}}$.

- **(b)** $\{-3, -2, -1, 0, 1, 2, 3\}$: Integers with absolute value at most 3: $\{x \in Z \mid -3 \leq x \leq 3\}$.
- **(c)** $\{m, n, o, p\}$: $\{x \mid x \text{ is a lowercase English letter from } m \text{ to } p \text{ inclusive}\}$.

3 Cardinality & Classification of Sets

3.1 Cardinality and the Empty Set

DEFINITION **Cardinality**

The **cardinality** of a finite set A, written $|A|$ or $n(A)$, is the number of distinct elements in A.

- For $A = \{1, 2, 3, ..., 10\}$, we have $|A| = 10$.
- A set with no elements has cardinality 0 and is called the **empty set** (or **null set**), written $\varnothing$ or $\{\}$:
  - $E = \{x \in N \mid x < 1\} = \varnothing$, so $|E| = 0$.
  - $\{x \mid x < x\} = \varnothing$, since no number is strictly less than itself.
  - $\{\text{humans who have walked on Mars}\} = \varnothing$.

PITFALL / WARNING **The Empty Set $\varnothing$ vs. The Singleton Set $\{\varnothing\}$**

It is easy to mix up $\varnothing$ and $\{\varnothing\}$:
- $\varnothing = \{\}$ is an empty box with **zero** elements ($|\varnothing| = 0$).
- $\{\varnothing\}$ is a box containing **one** element—namely, an empty box ($|\{\varnothing\}| = 1$, since $\varnothing \in \{\varnothing\}$).

3.2 Subsets and Proper Subsets

DEFINITION **Subset & Proper Subset**

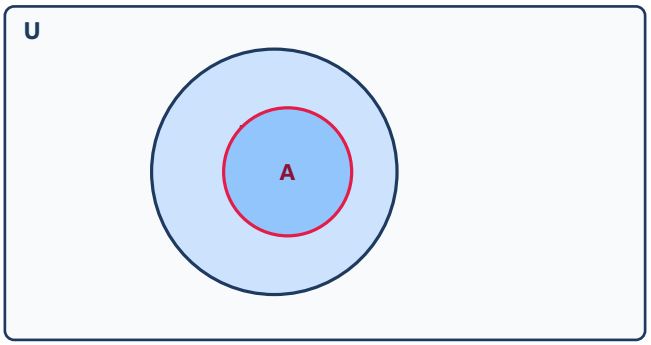
Let A and B be sets:

1. **Subset ($A \subseteq B$):** A is a **subset** of B if every element of A is also an element of B:

$$A \subseteq B \iff \forall x(x \in A \to x \in B)$$

   When $A \subseteq B$, we can also say B is a **superset** of A ($B \supseteq A$).
2. **Proper Subset ($A \subset B$):** A is a **proper subset** of B if $A \subseteq B$ and $A \neq B$ (meaning B has at least one element not in A).

For instance, $\{1, 2, 3\} \subset \{1, 2, 3, 4, 5\}$, and the standard number systems form a chain of subsets: $N \subseteq Z \subseteq Q \subseteq R \subseteq C$.



Venn diagram showing A as a subset of B (A ⊆ B) inside universe U.

CORE CONCEPT **Basic Subset Properties**

For any sets A, B, and C:

1. **Reflexivity:** $A \subseteq A$ (every set is a subset of itself).
2. **Empty Set:** $\varnothing \subseteq A$ (the empty set is a subset of every set).
3. **Transitivity:** If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

INTUITION & DEEP DIVE **Why Is $\varnothing$ a Subset of Every Set?**

Saying $\varnothing \subseteq A$ means $\forall x(x \in \varnothing \to x \in A)$. Because $\varnothing$ has no elements, the condition $x \in \varnothing$ is always false, making the implication vacuously true. Put another way: for $\varnothing \subseteq A$ to fail, we would need to find an element inside $\varnothing$ that is missing from A—which is impossible.

EXAMPLE & WALKTHROUGH **Finding Intermediate Proper Subsets**

Problem: Let $X = \{1, 2, 5\}$ and $Z = \{1, 2, 5, 7, 8\}$. Find all sets Y such that $X \subset Y$ and $Y \subset Z$.

Solution:

- Because $X \subset Y$ is a **proper** subset, Y must contain $1, 2, 5$ plus at least one more element.
- Because $Y \subset Z$ is also a **proper** subset, every element of Y must come from Z, and Y cannot equal all of Z.
- The remaining elements in Z are 7 and 8. Including just one of them gives the two valid possibilities:

$$Y_1 = \{1, 2, 5, 7\} \quad \text{and} \quad Y_2 = \{1, 2, 5, 8\}$$

3.3 Equal, Equivalent, and Disjoint Sets

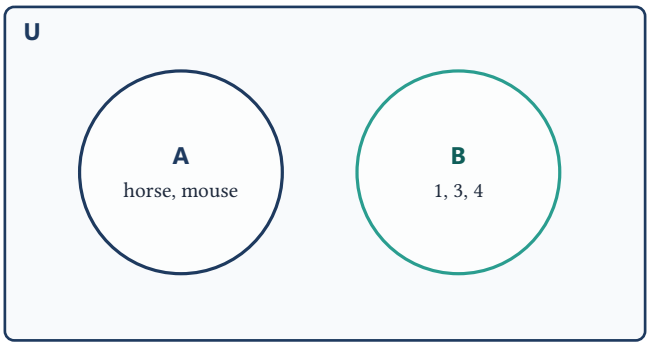
DEFINITION **Equal, Equivalent, and Disjoint Sets**

Let A and B be sets:

1. **Equal Sets ($A = B$):** A and B contain the exact same elements:

   $$A = B \iff (A \subseteq B) \land (B \subseteq A)$$

   Set equality is reflexive ($A = A$), symmetric ($A = B \implies B = A$), and transitive ($A = B$ and $B = C \implies A = C$). Order and repetition do not affect equality: $\{3, 2, 2, 3\} = \{2, 3\}$.
2. **Equivalent Sets ($A \sim B$):** A and B have the same cardinality ($|A| = |B|$). For example, $\{1, 2, 3\} \sim \{a, b, c\}$ because both have 3 elements, even though the sets are not equal.
3. **Disjoint Sets ($A \cap B = \varnothing$, or `A // B`):** A and B share no elements in common.



Disjoint sets A = {horse, mouse} and B = {1, 3, 4} with no elements in common.

3.4 Power Sets

DEFINITION **Power Set**

The **power set** of A, written $P(A)$ or 2^A, is the set of all subsets of A:

$$P(A) = \{S \mid S \subseteq A\}$$

If $|A| = n$, then $|P(A)| = 2^n$ (since each of the n elements can either be included or excluded when forming a subset).

- If $A = \{1, 2\}$, then $|P(A)| = 2^2 = 4$ and $P(A) = \{\varnothing, \{1\}, \{2\}, \{1, 2\}\}$.
- For $\{\varnothing\}$, we have $|\{\varnothing\}| = 1$, so $P(\{\varnothing\}) = \{\varnothing, \{\varnothing\}\}$ has $2^1 = 2$ elements.
- If $A = \{1, 2, 3\}$, then $P(A)$ has $2^3 = 8$ subsets, which we can list by size:

$$P(A) = \{\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

EXAMPLE & WALKTHROUGH **Working with Subsets and Venn Diagrams**

1. Determine whether the first set is a subset of the second, the second is a subset of the first, or neither:

- **(a)** People who speak English vs. people who speak English with an Australian accent: The second set is a proper subset of the first ($B \subset A$).
- **(b)** Fruits vs. citrus fruits: Every citrus fruit is a fruit, so the second set is a proper subset of the first ($B \subset A$).
- **(c)** Students studying discrete mathematics vs. students studying data structures: Neither is a subset of the other, since a student might take one course without taking the other.

2. Months without the letter R: Inside the universal set U of all 12 months, the subset M of months whose English names do not contain the letter `R` is:

$$M = \{\text{May}, \text{June}, \text{July}, \text{August}\}$$

The other 8 months (`January` through `April`, and `September` through `December`) lie outside M in $U - M$.

3. Visualizing subset relationships:

- If $A \subseteq B$ and $B \subseteq C$, we draw concentric circles A inside B inside C.
- If $A \subset B$ and $A \subset C$, circle A sits entirely inside the overlap region $B \cap C$ of circles B and C.
- **Proof that $A \subseteq B$ and $B \subseteq C \implies A \subseteq C$:** Take any $x \in A$. Since $A \subseteq B$, we have $x \in B$. Then, since $B \subseteq C$, we have $x \in C$. Thus every element of A is in C, so $A \subseteq C$. $\square$

4 Set Operations

Given sets A and B inside a universal set U, we can build new sets using several standard operations.

4.1 Intersection, Union, and Complement

DEFINITION **Intersection, Union, and Complement**

1. **Intersection ($A \cap B$):** Elements belonging to **both** A and B:

$$A \cap B = \{x \mid x \in A \land x \in B\}$$

   For $A = \{1, 5, 7, 9\}$ and $B = \{1, 2, 3\}$, $A \cap B = \{1\}$.
2. **Union ($A \cup B$):** Elements belonging to A **or** B (or both):

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

For $A = \{1, 2\}$ and $B = \{2, 5, 6\}$, $A \cup B = \{1, 2, 5, 6\}$.

3. **Complement ($\overline{A}$ or A^c):** Elements in the universe U that are **not** in A:

$$\overline{A} = \{x \mid x \in U \wedge x \notin A\} = U - A$$

If $U = \{1, 2, ..., 10\}$ and $A = \{1, 4, 5\}$, then $\overline{A} = \{2, 3, 6, 7, 8, 9, 10\}$.

EXAMPLE & WALKTHROUGH **Translating Everyday Statements into Set Notation**

Let A be the set of red fruits, B be the set of imported fruits, and E be the set of fruits sold at Store X. How do we write **"the set of fruits sold at Store X that are red or imported"**?

- Being "red or imported" means belonging to $A \cup B$.
- Also being sold at Store X means belonging to E as well.
- Combining the two gives:

$$E \cap (A \cup B) \quad \text{(or equivalently, } (E \cap A) \cup (E \cap B))$$

4.2 Set Difference and Symmetric Difference

DEFINITION **Difference & Symmetric Difference**

1. **Difference ($A - B$ or $A \setminus B$):** Elements in A that are **not** in B:

$$A - B = \{x \mid x \in A \wedge x \notin B\} = A \cap \overline{B}$$

If $A = \{1, 2, ..., 10\}$ and $B = \{1, 2, 3, 11\}$, then $A - B = \{4, 5, ..., 10\}$ (while $B - A = \{11\}$).

2. **Symmetric Difference ($A \oplus B$):** Elements in A or B, but **not both** (the set version of logical `XOR`):

$$A \oplus B = (A \cup B) - (A \cap B) = (A - B) \cup (B - A)$$

If $A = \{1, 2, 3\}$ and $B = \{3, 5, 6\}$, then $A \oplus B = \{1, 2, 5, 6\}$.

Symmetric difference is both **commutative** ($A \oplus B = B \oplus A$) and **associative** ($(A \oplus B) \oplus C = A \oplus (B \oplus C)$).

4.3 Cartesian Product

DEFINITION **Cartesian Product**

The **Cartesian product** of A and B, written $A \times B$, is the set of all **ordered pairs** (a, b) with $a \in A$ and $b \in B$:

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$$

For example, if $A = \{1, 2, 3\}$ and $B = \{a, b\}$, then:

$$A \times B = \{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)\}$$

Key properties to keep in mind:

- **Cardinality:** For finite sets, $|A \times B| = |A| \cdot |B|$.
- **Order matters:** $(a, b) \neq (b, a)$ unless $a = b$. Consequently, $A \times B \neq B \times A$ in general (unless $A = B$ or one of the sets is empty).
- **Empty set:** If $A = \varnothing$ or $B = \varnothing$, then $A \times B = \varnothing$.

EXAMPLE & WALKTHROUGH **Three-Way Cartesian Product**

Let $A = \{1, 2\}$, $B = \{x, y, z\}$, and $C = \{3, 4\}$. Then $|A \times B \times C| = 2 \times 3 \times 2 = 12$, and the ordered triples in $A \times B \times C$ are:

$$A \times B \times C = \{(1, x, 3), (1, x, 4), (1, y, 3), (1, y, 4), (1, z, 3), (1, z, 4),$$
$$(2, x, 3), (2, x, 4), (2, y, 3), (2, y, 4), (2, z, 3), (2, z, 4)\}$$

EXAMPLE & WALKTHROUGH **Set Operations Practice**

1. Interpreting set expressions: Let A be students who live within one mile of school, and B be students who walk to classes.

- $A \cap B$: Students who live within one mile of school **and** walk to classes.
- $A \cup B$: Students who live within one mile of school **or** walk to classes.
- $A - B$: Students who live within one mile of school but do **not** walk to classes.
- $B - A$: Students who walk to classes but live more than one mile from school.

2. Writing set expressions: Let A be sophomores and B be students taking discrete mathematics.

- Sophomores taking discrete math: $\boldsymbol{A \cap B}$.
- Sophomores not taking discrete math: $\boldsymbol{A - B}$ (or $A \cap \overline{B}$).
- Students who are sophomores or taking discrete math: $\boldsymbol{A \cup B}$.
- Students who are either not sophomores or not taking discrete math: $\boldsymbol{\overline{A} \cup \overline{B}}$ (which equals $\overline{A \cap B}$).

3. Computing set operations:

- For $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$: $A \cup B = \{0, 1, 2, 3, 4, 5, 6\}$, $A \cap B = \{3\}$, $A - B = \{1, 2, 4, 5\}$, $B - A = \{0, 6\}$.
- For $A = \{a, b, c, d, e\}$ and $B = \{a, b, c, d, e, f, g, h\}$ (note $A \subset B$): $A \cup B = B$, $A \cap B = A$, $A - B = \varnothing$, $B - A = \{f, g, h\}$.

5 The Principle of Inclusion–Exclusion

When two finite sets overlap, adding $|A| + |B|$ counts every element in $A \cap B$ twice. Subtracting that overlap once gives the Principle of Inclusion–Exclusion.

CORE CONCEPT **Principle of Inclusion–Exclusion**

For finite sets A, B, C:

1. **Two Disjoint Sets ($\boldsymbol{A \cap B = \varnothing}$):** $|A \cup B| = |A| + |B|$
2. **Two General Sets:** $|A \cup B| = |A| + |B| - |A \cap B|$
3. **Three General Sets:**

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

INTUITION & DEEP DIVE **Why Do We Add Back |A ∩ B ∩ C|?**

Think about an element x that belongs to all three sets A, B, and C:

- It is counted 3 times in $|A| + |B| + |C|$.
- It is subtracted 3 times in $-|A \cap B| - |A \cap C| - |B \cap C|$, bringing its net count down to 0.
- Adding $+|A \cap B \cap C|$ adds it back once so that x is counted **exactly once** in the union.

EXAMPLE & WALKTHROUGH **Three-Set Survey Problem**

Problem: In a survey of 60 people ($|U| = 60$) regarding three magazines (E, G, and F):

- $|E| = 25$, $|G| = 26$, and $|F| = 26$.
- $|E \cap F| = 9$, $|E \cap G| = 11$, and $|G \cap F| = 8$.
- 8 people read none of the three magazines ($|\overline{E \cup G \cup F}| = 8$).

How many people read all three magazines?

Solution:

1. Since 8 of the 60 people read none of the magazines, the number who read at least one is:
$$|E \cup G \cup F| = 60 - 8 = 52$$
2. Plugging the known counts into the 3-set Inclusion–Exclusion formula:
$$52 = (25 + 26 + 26) - (11 + 9 + 8) + |E \cap G \cap F|$$
$$52 = 77 - 28 + |E \cap G \cap F| = 49 + |E \cap G \cap F|$$
$$|E \cap G \cap F| = 52 - 49 = 3$$

So **3 people** read all three magazines.

6 Laws of Set Algebra and Duality

Set operations follow algebraic rules that mirror propositional logic ($\cup$ acts like $\vee$, $\cap$ acts like $\wedge$, complement acts like $\neg$, $\varnothing$ acts like False, and U acts like True). Because of this symmetry, every law comes with a **dual**: swapping $\cup \leftrightarrow \cap$ and $\varnothing \leftrightarrow U$ turns any valid set identity into another valid identity.

Law Name	Primary Identity	Dual Identity
1. Identity Laws	$A \cup \varnothing \equiv A$	$A \cap U \equiv A$
2. Null / Domination Laws	$A \cap \varnothing \equiv \varnothing$	$A \cup U \equiv U$
3. Complement Laws	$A \cup \overline{A} \equiv U$	$A \cap \overline{A} \equiv \varnothing$
4. Idempotent Laws	$A \cup A \equiv A$	$A \cap A \equiv A$
5. Involution Law	$\overline{(\overline{A})} \equiv A$	(Self-dual)
6. Absorption Laws	$A \cup (A \cap B) \equiv A$	$A \cap (A \cup B) \equiv A$
7. Commutative Laws	$A \cup B \equiv B \cup A$	$A \cap B \equiv B \cap A$
8. Associative Laws	$A \cup (B \cup C) \equiv (A \cup B) \cup C$	$A \cap (B \cap C) \equiv (A \cap B) \cap C$
9. Distributive Laws	$A \cup (B \cap C) \equiv (A \cup B) \cap (A \cup C)$	$A \cap (B \cup C) \equiv (A \cap B) \cup (A \cap C)$
10. De Morgan's Laws	$\overline{A \cap B} \equiv \overline{A} \cup \overline{B}$	$\overline{A \cup B} \equiv \overline{A} \cap \overline{B}$
11. 0/1 Laws	$\overline{\varnothing} \equiv U$	$\overline{U} \equiv \varnothing$

7 Proving Set Identities

There are three common ways to check or prove that two set expressions always produce the same set.

7.1 Venn Diagrams

To see why $A \cap (B \cup C) \equiv (A \cap B) \cup (A \cap C)$, we can draw two 3-circle Venn diagrams:

- On the left, shade $B \cup C$ and keep only the part inside A.
- On the right, shade $A \cap B$ and $A \cap C$ and combine them.

Both give the exact same shaded region. Venn diagrams are great for building intuition, while membership tables and algebraic proofs provide formal verification.

7.2 Membership Tables

Just like truth tables in logic, a **membership table** checks every possible combination of whether an element x is inside (`1`) or outside (`0`) each base set. Two expressions are equal if their columns match in every row:

A	B	C	$B \cup C$	$\boldsymbol{A \cap (B \cup C)}$	$A \cap B$	$A \cap C$	$\boldsymbol{(A \cap B) \cup (A \cap C)}$
0	0	0	0	**0**	0	0	**0**
0	0	1	1	**0**	0	0	**0**
0	1	0	1	**0**	0	0	**0**
0	1	1	1	**0**	0	0	**0**
1	0	0	0	**0**	0	0	**0**
1	0	1	1	**1**	0	1	**1**
1	1	0	1	**1**	1	0	**1**
1	1	1	1	**1**	1	1	**1**

Since the two bold columns match across all 8 rows, $A \cap (B \cup C) \equiv (A \cap B) \cup (A \cap C)$.

7.3 Algebraic Proofs

We can also rewrite one side step-by-step using the laws of set algebra (plus the difference rule $X - Y \equiv X \cap \overline{Y}$).

For example, to prove $(A \cap B) \cup \left(A \cap \overline{B}\right) \equiv A$:

$$\begin{aligned}
(A \cap B) \cup \left(A \cap \overline{B}\right) &\equiv A \cap \left(B \cup \overline{B}\right) && \text{(Distributive Law)} \\
&\equiv A \cap U && \text{(Complement Law)} \\
&\equiv A && \text{(Identity Law)} \quad \square
\end{aligned}$$

EXAMPLE & WALKTHROUGH **Algebraic Proofs with Set Difference**

1. Prove that $A \cap (B - C) \equiv (A \cap B) - (A \cap C)$:

Starting from the right-hand side:

$$\begin{aligned}
(A \cap B) - (A \cap C) &\equiv (A \cap B) \cap \overline{(A \cap C)} && \text{(Definition of Difference)} \\
&\equiv (A \cap B) \cap \left(\overline{A} \cup \overline{C}\right) && \text{(De Morgan's Law)} \\
&\equiv \left((A \cap B) \cap \overline{A}\right) \cup \left((A \cap B) \cap \overline{C}\right) && \text{(Distributive Law)} \\
&\equiv \left(B \cap \left(A \cap \overline{A}\right)\right) \cup \left(A \cap \left(B \cap \overline{C}\right)\right) && \text{(Commutative \& Associative Laws)} \\
&\equiv (B \cap \varnothing) \cup \left(A \cap \left(B \cap \overline{C}\right)\right) && \text{(Complement Law)} \\
&\equiv \varnothing \cup \left(A \cap \left(B \cap \overline{C}\right)\right) && \text{(Domination Law)} \\
&\equiv A \cap \left(B \cap \overline{C}\right) && \text{(Identity Law)} \\
&\equiv A \cap (B - C) && \text{(Definition of Difference)} \quad \square
\end{aligned}$$

2. Prove that $(A \cup B) - C \equiv (A - C) \cup (B - C)$:

Starting from the left-hand side:

$$
\begin{aligned}
(A \cup B) - C &\equiv (A \cup B) \cap \overline{C} && \text{(Definition of Difference)} \\
&\equiv \overline{C} \cap (A \cup B) && \text{(Commutative Law)} \\
&\equiv \left(\overline{C} \cap A\right) \cup \left(\overline{C} \cap B\right) && \text{(Distributive Law)} \\
&\equiv \left(A \cap \overline{C}\right) \cup \left(B \cap \overline{C}\right) && \text{(Commutative Law)} \\
&\equiv (A - C) \cup (B - C) && \text{(Definition of Difference)} \quad \square
\end{aligned}
$$